

Section 2: System Design Fundamentals

Forward Proxy Vs. Reverse Proxy

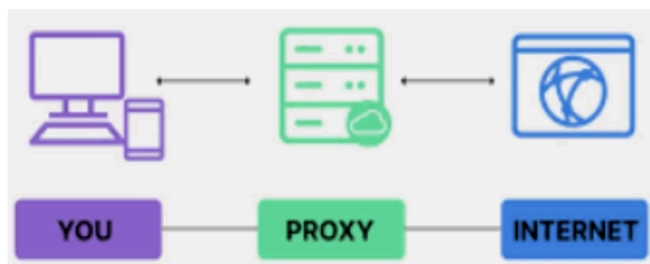
Proxies play a crucial role in modern systems by enhancing security, improving performance and managing traffic efficiently. However, not all proxies work the same way. Some protect clients for other safeguards servers.

In this session, we are going to break down the key differences between forward proxies and reverse proxies. We will explore the real world use cases and understand when to use each of them in the system design.

By the end of this session, you will have a solid understanding of how proxies function and how they fit into scalable, secure and high performance architectures.

What is a Proxy?

A proxy server is the intermediary that sits between the client, which is a browser or an app on your device, and the server. Instead of the client communicating directly with the server, the proxy relies on the client to send the request to the proxy and then it relays this request to the server. And when the response comes from the server, it will get that response and send that response back to the client.



Proxies are widely used in the system architecture for several reasons:

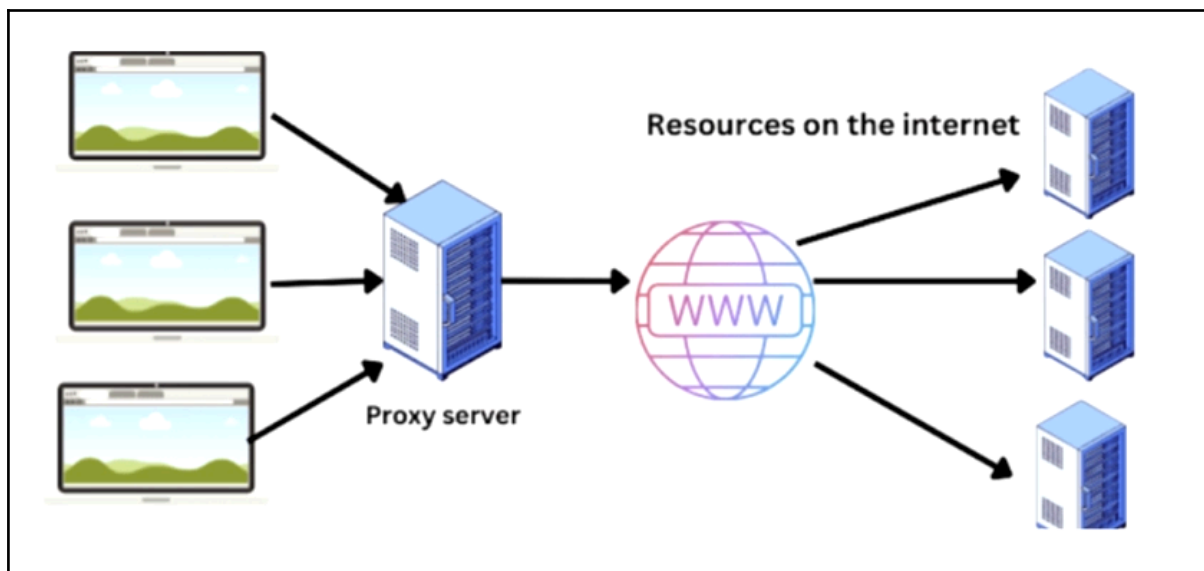
1. **Security:** It helps in protecting both clients and servers from threats.
2. **Caching:** They store frequently accessed content, which eventually reduces latency and improves the overall performance.
3. **Traffic Control:** They distribute requests efficiently, preventing the overloads of the backend systems.
4. **Anonymity:** Sometimes these proxies mask the IP addresses of the client and help maintain the user's privacy.

There are mainly two types of proxies, and each serves a different purpose.

- **Forward Proxy:** It is actually used to protect and manage the client request. The forward proxy always sits in front of the client.
- **Reverse Proxy:** It is used to manage and optimize server responses.

What is a Forward Proxy?

A forward proxy acts as an intermediary between clients and the internet. Instead of a client directly accessing the website or an online service, it first sends its request to the forward proxy, and the proxy then forwards this request to the destination server. The proxy will retrieve the response and send it back to the client.



Advantages of forward proxies are:

1. **Content Filtering:** Many organizations use forward proxies to restrict access to certain websites or harmful content which are not supposed to be opened from their internal network.
2. **Anonymity:** The proxies hide the client's IP address, which makes it very useful for anonymous browsing through VPNs or Tor networks.
3. **Caching:** Frequently accessed pages and resources can be stored in these proxies, which will reduce the load time and it will actually save the bandwidth a lot.

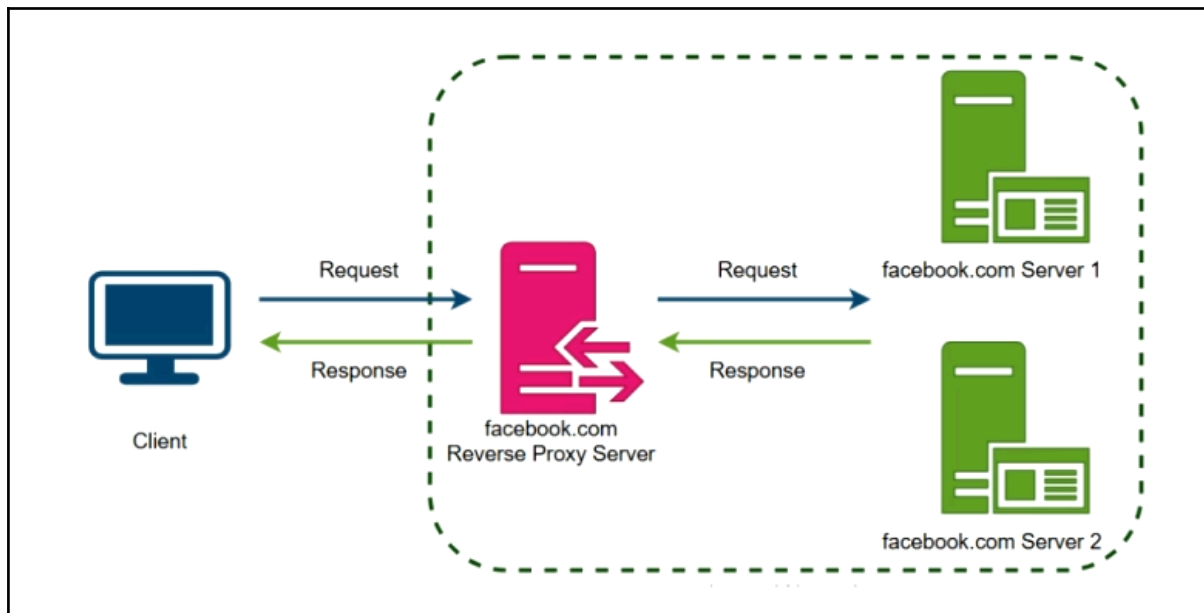
Some common use cases of forward proxies include:

- **Bypassing the Geo-restrictions:** Users that are restricted from accessing some content in a region can access blocked content, by routing traffic through a proxy that is in a different region.
- **Anonymous Browsing:** VPN services and Tor networks allow the users to browse the internet without revealing their actual identity.
- **Performance Optimization:** Companies use forward proxies to cache these web pages, which will improve the overall loading speed and it will reduce the bandwidth usage.

In simple terms, a forward proxy is mainly used by the clients to control and optimize their access to the internet.

What is a Reverse Proxy?

Unlike a forward proxy that sits in front of the client, a reverse proxy sits in front of the backend server and handles requests on their behalf. Clients don't directly communicate to the backend server, but all the requests from the clients actually go to the reverse proxy first.



The reverse proxy is a very powerful tool. It is used in system architecture for many reasons like:

- **Load Balancing:** It distributes incoming requests across multiple backend servers, preventing overload of any single server and keeping the performance smooth for the application.
- **Caching:** Frequently requested content is stored in this proxy, which reduces the response time and improves the overall performance of the application.
- **Security:** It also acts like a shield, hiding the actual backend server and filtering out the malicious traffic.
- **SSL Termination:** The reverse proxy handles encryption and decryption of the HTTP traffic, offloading this heavy task from the backend server.

In summary, a reverse proxy is used by servers to optimize performance, security and scalability. It plays a critical role in modern web architecture. It ensures that applications can handle millions of requests efficiently.

Key Differences - Forward Vs. Reverse Proxy

<i>Feature</i>	<i>Forward Proxy</i>	<i>Reverse Proxy</i>
Position	It sits in front of the clients, handling the outgoing requests from users to the internet.	It sits in front of the backend server, managing incoming requests from users to servers.
Who uses it?	Forward proxies are used by clients, individual users and organizations to provide anonymity and security.	Reverse proxies are used by servers, web applications, APIs and backend infrastructure.
Purpose	Privacy, content filtering and bypassing the restrictions.	Load balancing, caching, security and optimizing the overall performance.
Example	VPN, Web Proxy	Nginx, CloudFlare, and AWS load balancers.

In summary, both forward and reverse proxies act as intermediaries but they serve very different roles in the network architecture. Forward proxies protect and manage client requests, whereas reverse proxies optimize and secure the network backend services.

Wrap-Up & What's Next?

- Key Takeaways:
 - A forward proxy helps clients interact with the internet securely.
 - A reverse proxy protects and optimizes backend servers.
 - Understanding proxy usage helps in designing scalable & secure architectures.
- Next Topic:
 - Load Balancers

Interview Questions

- Basic Concepts
 - What is a proxy server, and why is it used?
 - Explain the key differences between a forward proxy and a reverse proxy.
- Technical Deep Dive
 - How does a forward proxy improve security and privacy?
 - How does a reverse help in load balancing and caching?
 - What are some real-world examples of forward and reverse proxies?
- Implementation & Use Cases
 - When should you use a forward proxy vs. a reverse proxy in system architecture?
 - What are some common tools and technologies used for each type of proxy?
 - How does a reverse proxy protect backend servers from DDoS attacks?
- Advanced Topics
 - How does SSL termination work in reverse proxy?
 - What are the advantages of using Cloudflare, Nginx, or HAProxy as a reverse proxy?
 - How does a transparent proxy differ from a forward or reverse proxy?